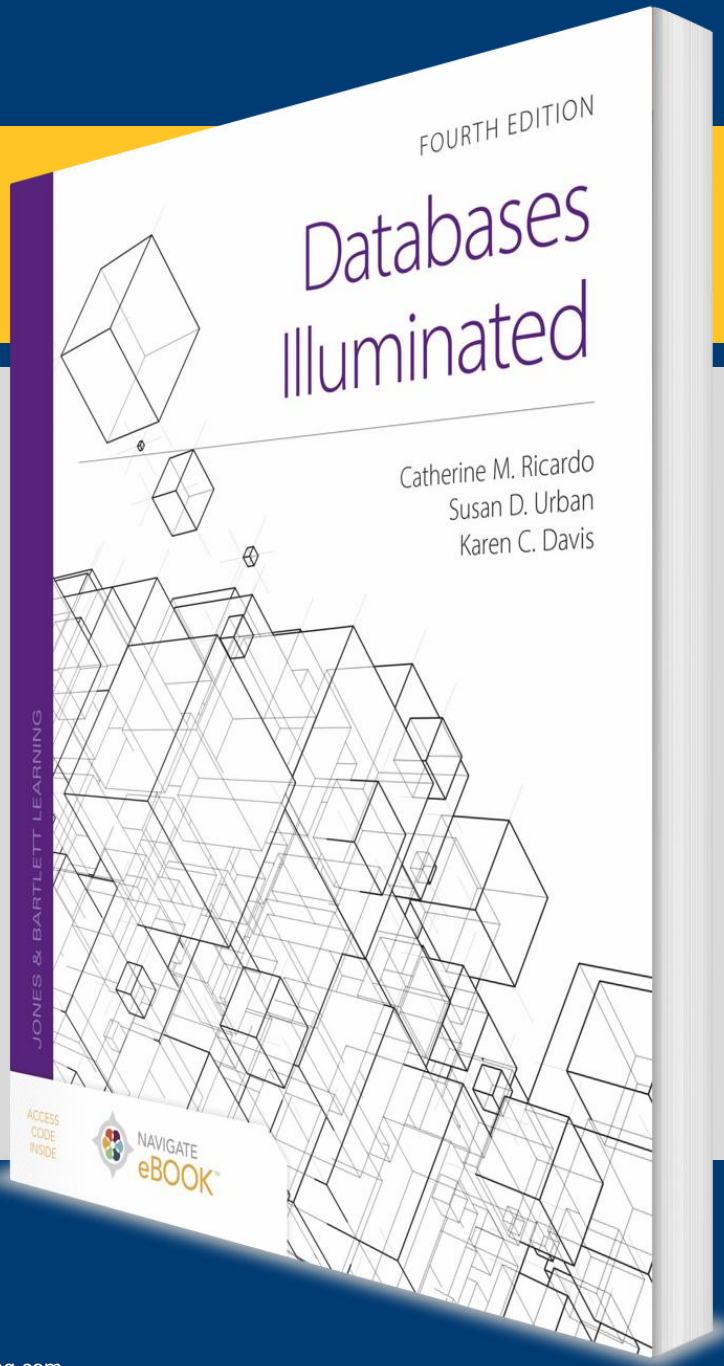
CHAPTER 3

The Entity Relationship Model



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Purpose of ER Model

- Facilitates database design
- Express logical properties of mini-world of interest within enterprise
- Conceptual level model
- Not limited to any particular DBMS
- ER diagrams used as design tools
- A semantic model – captures meanings
- Consists of entities, attributes, relationships

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Basic Symbols for ER Diagram

- Entity – rectangle
- Attribute – oval
- Relationship – diamond
- Link – line

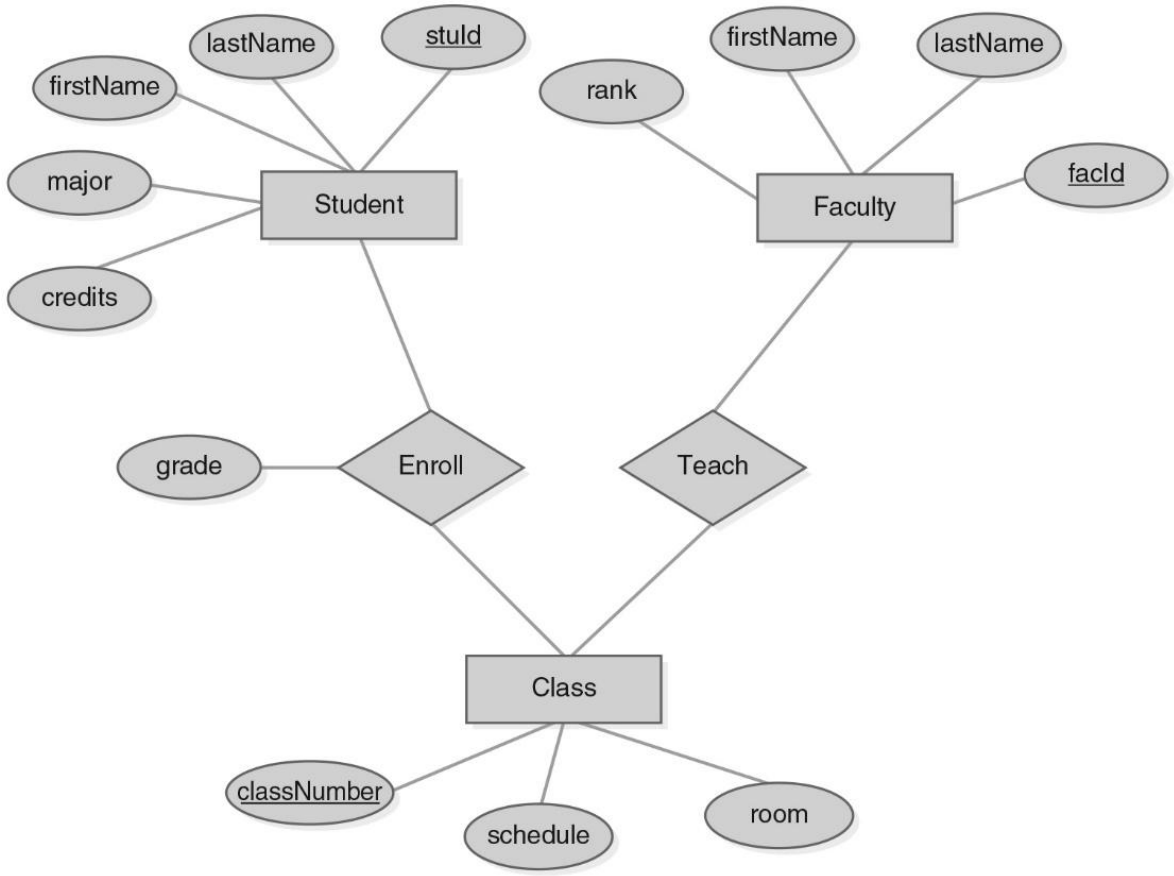

Figure 3.1A Simplified ER Diagram for the University Database

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Entity

- Object that exists and can be distinguished from other objects
- Can be person, place, event, object, concept in the real world
- Can be physical object or abstraction
- Entity **instance** is a particular person, place, etc.
- Entity **type** is a category of entities
- Entity **set** is a collection of entities of same type – must be well-defined
- Entity type forms **intension** of entity – permanent definition part
- Entity instances form **extension** of entity – all instances that fulfill the definition at the most recent point in time
- In ER diagram, rectangle represents entity set, not individual entities

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Attributes

- Defining properties or qualities of entity type
- Represented by ovals on ER diagram
- **Domain** – set of allowable values for attribute
- Attribute maps entity set to domain
- May have **null values** for some entity instances – no mapping to domain for those instances
- May be **multi-valued** – use double oval on ER diagram
- May be **composite** – use oval for composite attribute, with ovals for components
- May be **derived** – use dashed oval

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Keys

- **Superkey**: attribute or set of attributes that uniquely identifies an entity
- **Composite key**: key with more than one attribute
- **Candidate key**: superkey such that no proper subset of its attributes is also a superkey (minimal superkey – no unnecessary attributes)
- **Primary key**: candidate key used as main way to identify entities and access records of entity set

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Relationships

- Connections or interactions between entity sets
- Represented by diamond on ER diagram
- Relationship **type** – category of relationships
- Relationship **set** – collection of relationships of same type, consists of relationship instances – relationships that exist at a given point in time
- Type forms **intension**; set forms **extension** of relationship
- Relationship can have descriptive attributes
- **Degree** of relationship – number of entity sets linked by the relationship
  - **Binary** – links two entity sets; set of ordered pairs
  - **Ternary** – links three entity sets; ordered triples
  - **N-ary** – links n entity sets; ordered n-tuples
  - **Unary** (**recursive**) – links entity set to itself; ordered tuples from same set

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Cardinality of Binary Relationships

- Number of entity instances to which another entity instance can map under the relationship at a given time
- **One-to-one**: X:Y is 1:1 if each entity in X is associated with at most one entity in Y and each entity in Y with at most one entity in X
- **One-to-many**: X:Y is 1:M if each entity in X can be associated with many entities in Y, but each entity in Y with at most one entity in X (many=more than one)
- **Many-to-many**: X:Y is M:M if each entity in X can be associated with many entities in Y, and each entity in Y with many entities in X
- **Figure 3.7** shows several representation methods
- Can use M, m, N, or n for “many”
- Can use a constant integer for “many” if there is some maximum number of entities for the relationship

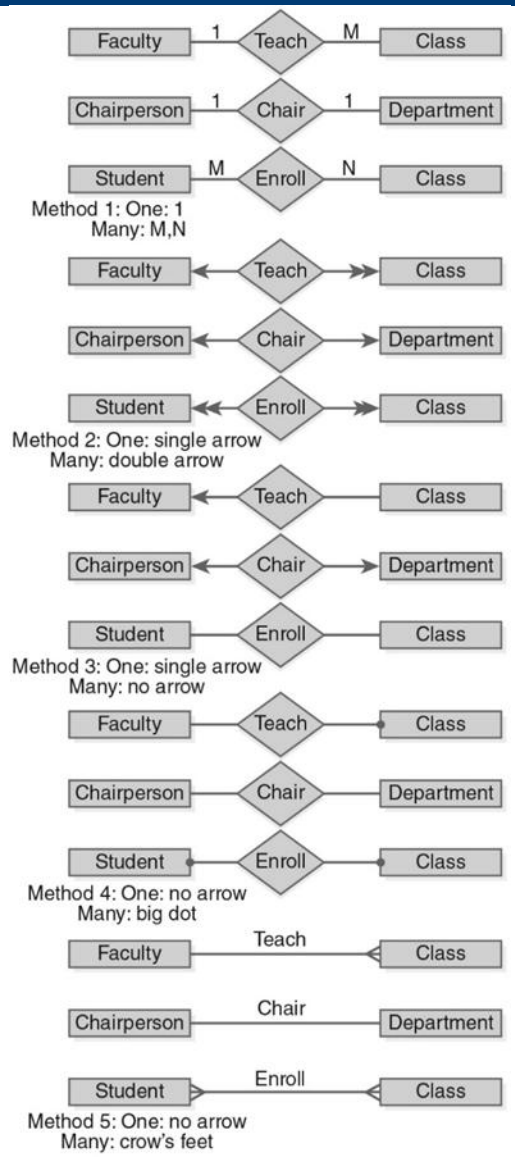

Figure 3.7 Showing Cardinalities on an ER Diagram

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- **Total participation**
  - Every member of entity set **must** participate in the relationship
  - Represented by double line from entity rectangle to relationship diamond
- **Partial participation**
  - Not every entity instance must participate
  - Represented by single line from entity rectangle to relationship diamond

See Figure 3.9

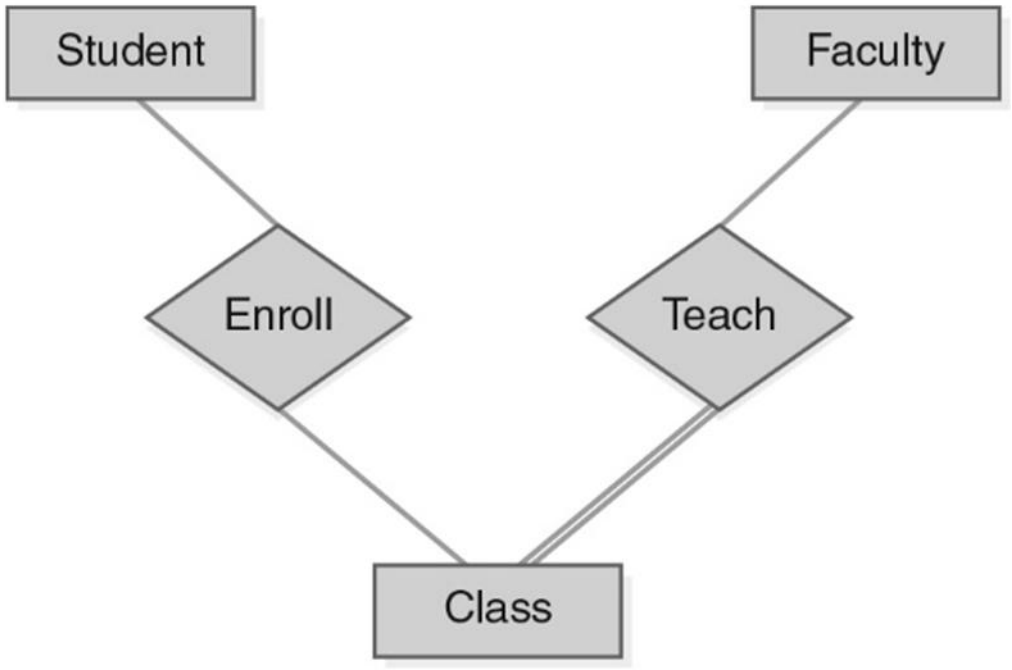

Figure 3.9 Total vs. Partial Participation in a Relationship

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(*min,max*) Notation for Cardinality and Participation

- ***min***
  - Least number of relationship instances an entity instance **must** participate in
  - Can be 0 (partial participation)
  - Or 1 or more (total participation)
- ***max***
  - Greatest number of relationship instances the entity **can** participate in
  - Can be 1
  - Or many (written M, m, N, n, or *)
  - Or some constant integer
- Written on line connecting entity rectangle to relationship diamond

See **Figure 3.10**

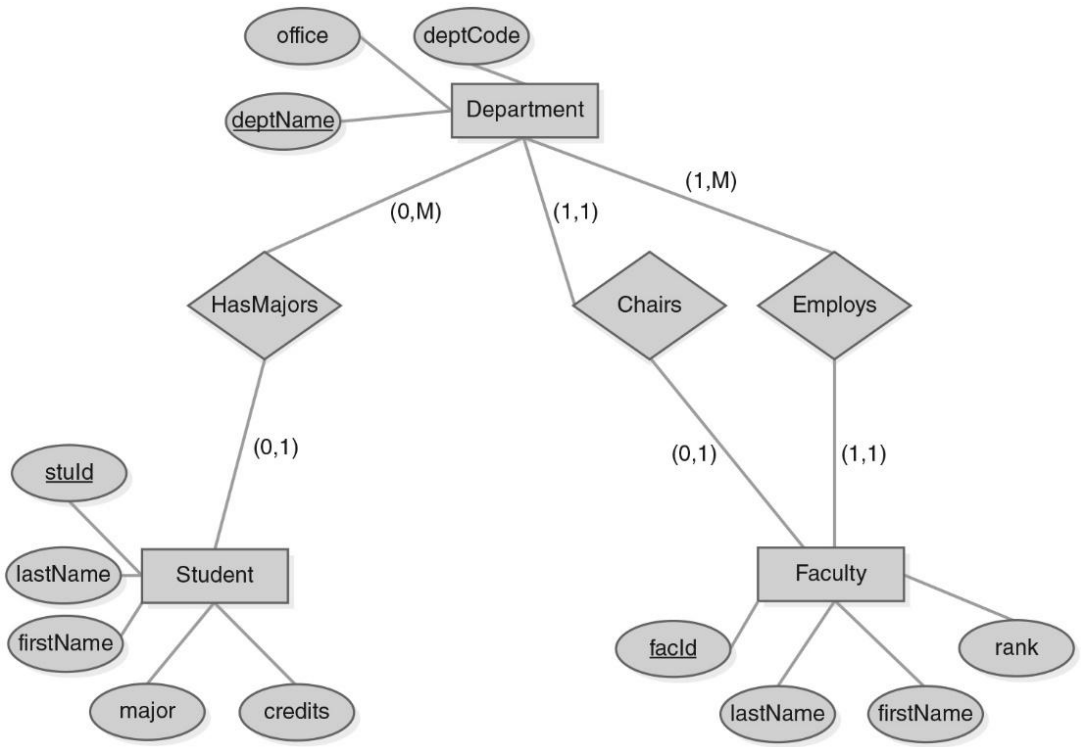

FIGURE 3.10
Part of the University ER Diagram with (*min, max*) Notation

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Roles

- **Role**: function that an entity plays in a relationship
- Optional to name role of each entity, but helpful in cases of
  - Recursive relationship – entity set relates to itself
  - Multiple relationships between same entity sets

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Existence Dependency and Weak Entities

- Entity Y is **existence dependent** on entity X is each instance of Y must have a corresponding instance of X
- Y must have **total participation** in its relationship with X
- **If Y does not have its own candidate key**, Y is called a **weak entity**, and X is **strong entity**
- Weak entity may have a partial key, a **discriminator**, that distinguishes instances of the weak entity that are related to the same strong entity
- Use double rectangle for weak entity, with double diamond for relationship connecting it to its strong entity
- Note: Not all existence dependent entities are weak – the lack of a key is essential to definition of weakness

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Examples

- University Database

See Figure 3.13

- Horse Racing Database

See Figure 3.14

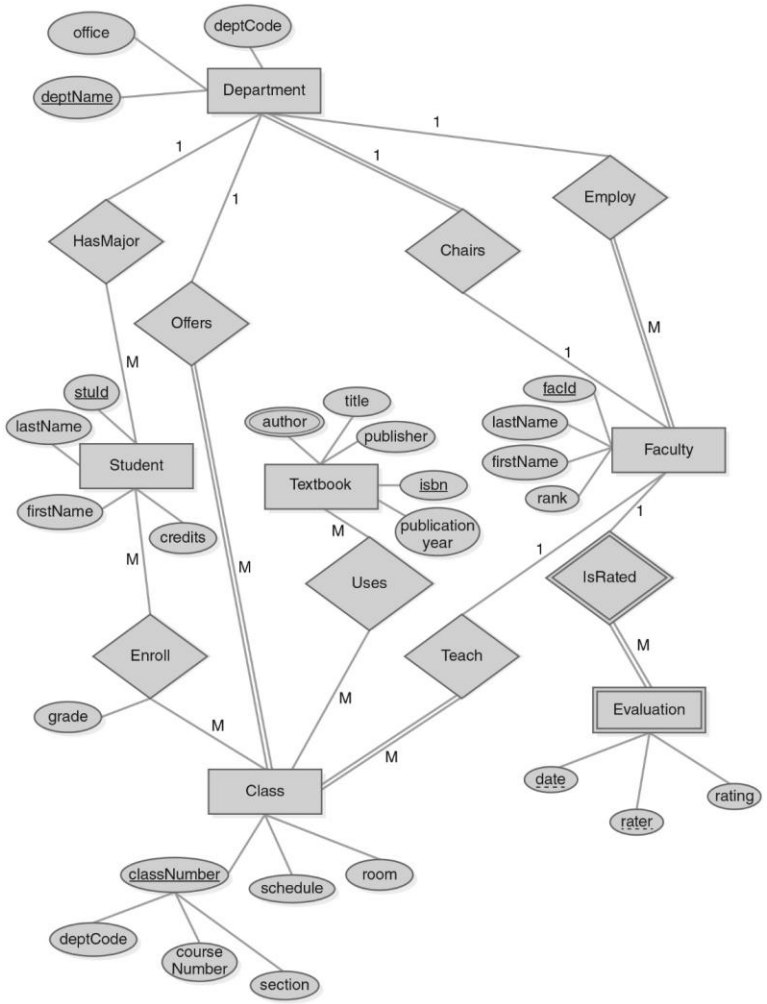

Figure 3.13 An ER Diagram for the University. Database

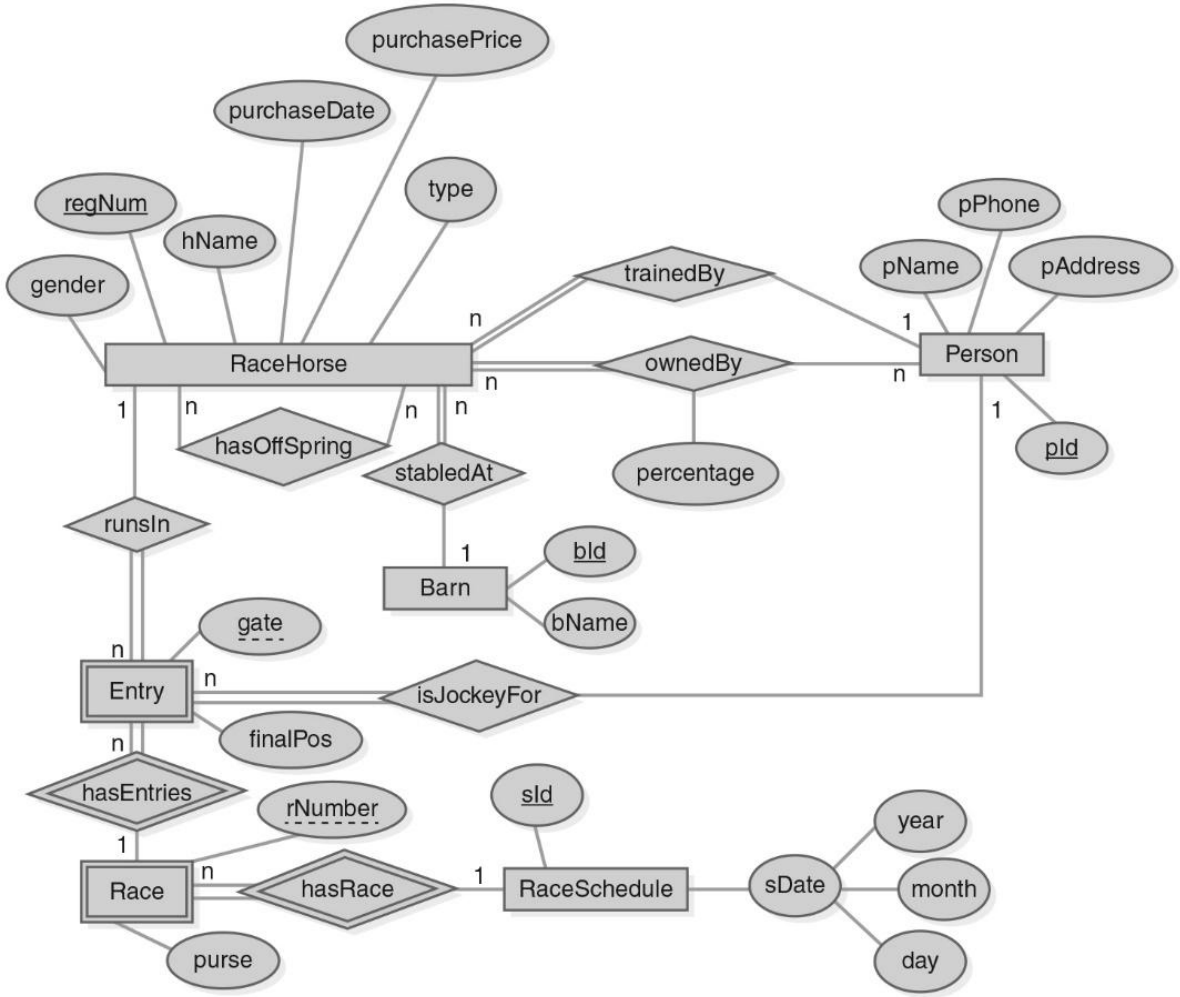

Figure 3.14 ER Diagram for `Horse Racing` **Database**

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Why Extend the ER Model?

- ER is suitable for traditional business applications
- ER is not semantically rich enough for advanced applications
- Applications where ER is inadequate
  - Geographic information systems
  - Search engines
  - Data mining
  - Multimedia
  - CAD/CAM
  - Software development
  - Engineering design
  - Financial services
  - Digital publishing
  - Telecommunications
  - ...and others

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Specialization Abstraction

- Specialization—needed when an entity set has subsets that have special attributes or that participate in special relationships
- Process of breaking up a set into subsets

  Ex: `Faculty` contains `AdjunctFac` and `FullTimeFac`
  - All `Faculty` have attributes `facId`, `lastName`, `firstName`, `rank`.
  - `AdjunctFac` also have `coursePayRate`
  - `FullTimeFac` have `annualSalary`

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Representing Specialization

- EER diagram shows specialization circle (*isa* relationship), and inheritance symbol (subset symbol)

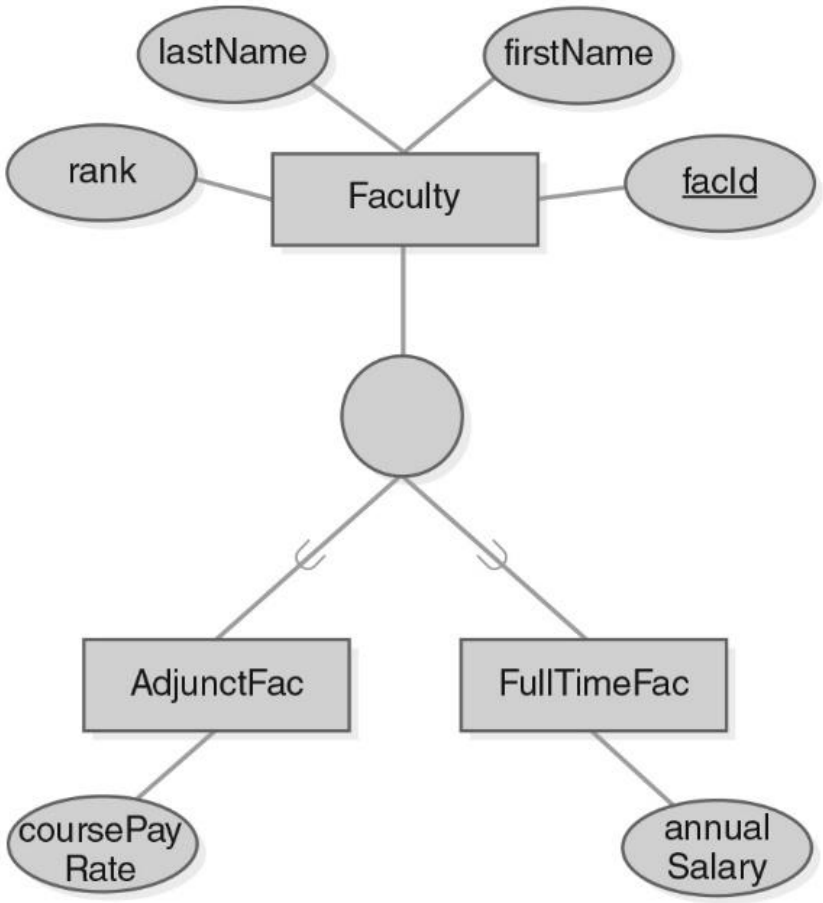

Figure 3.15A Specialization with Two Subsets

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- Specialization can also involve just one subset – no need for circle, but show inheritance symbol

Ex. `Class` has `LabClass` specialization

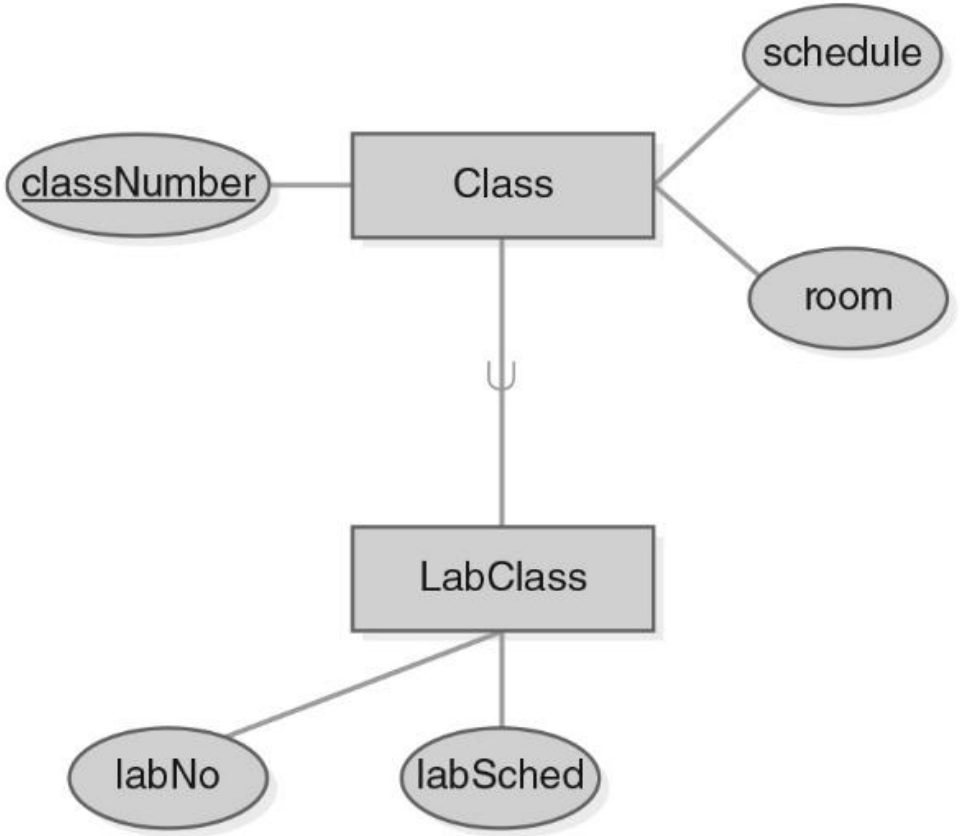

Figure 3.15B Specialization with One Subset

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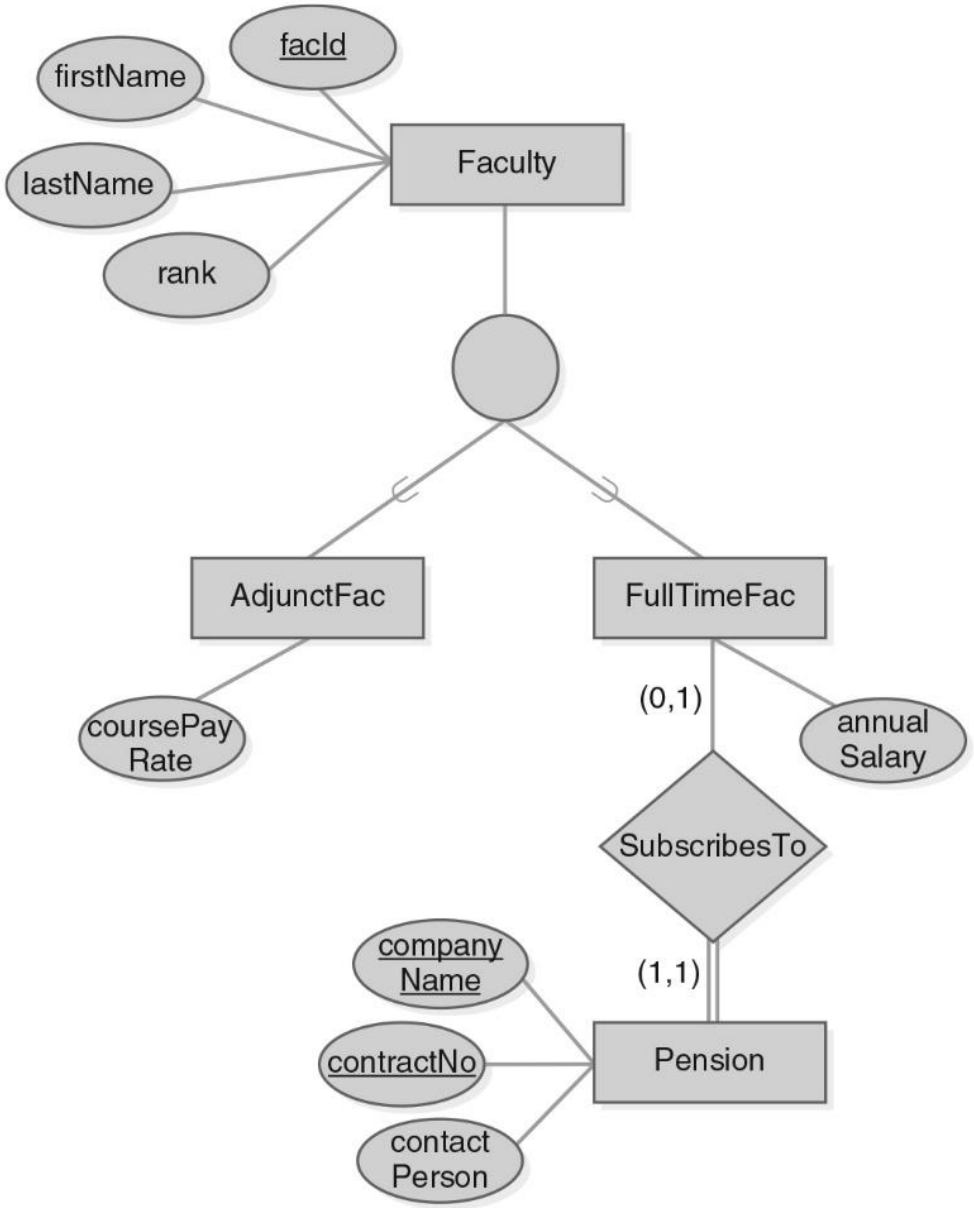
- Subsets can participate in their own relationships

Ex. `FullTimeFac subscribes To Pension`

Figure 3.15C Subset with Relationship



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Generalization Abstraction

- Inverse of specialization
- Recognizing that sets have common properties and identifying a superset for them
  Ex. `Student` and `Faculty` are both people
- Bottom-up process, as opposed to top-down process of specialization
- EER diagram is the same as for specialization

See Figure 3.15D

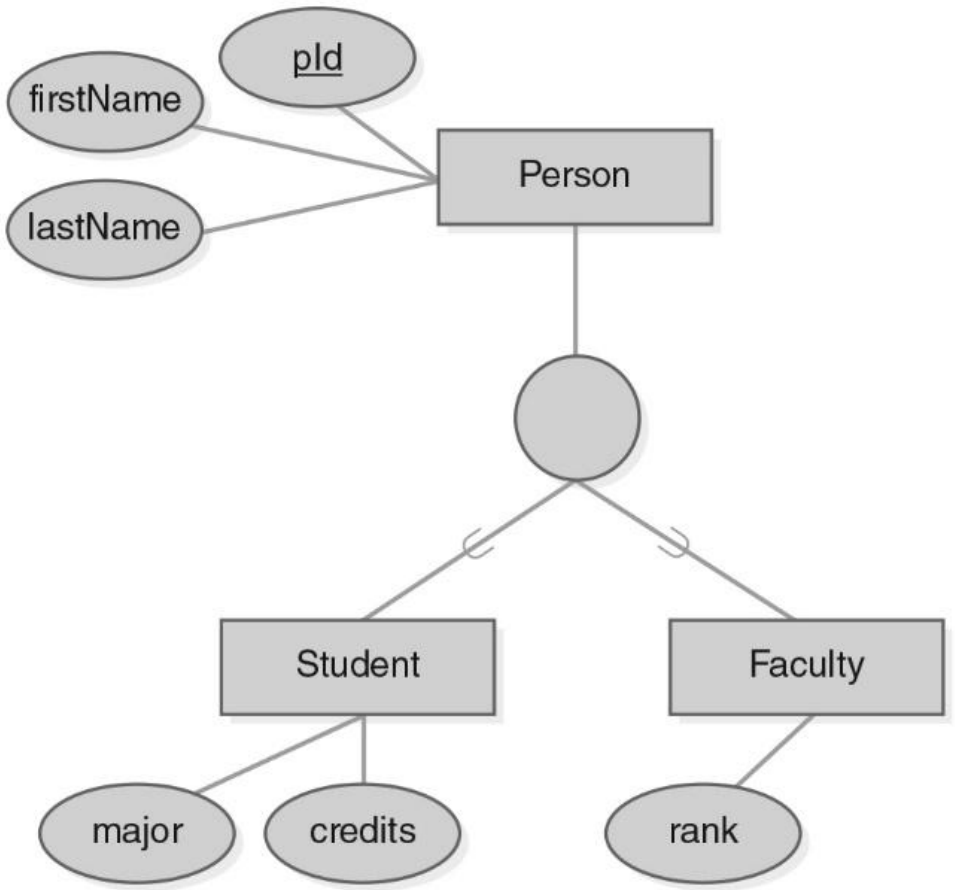

Figure 3.15D Generalization with `Person` Superset

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Generalization/Specialization Constraints

- Subsets can be **overlapping** or **disjoint**
- Place **o** or **d** in specialization circle to indicate constraint
- Specialization can be **total** (every member of superset must be in some subset) or **partial**
  - Total-double line connecting superset to specialization circle
  - Partial-single line

See Figure 3.16

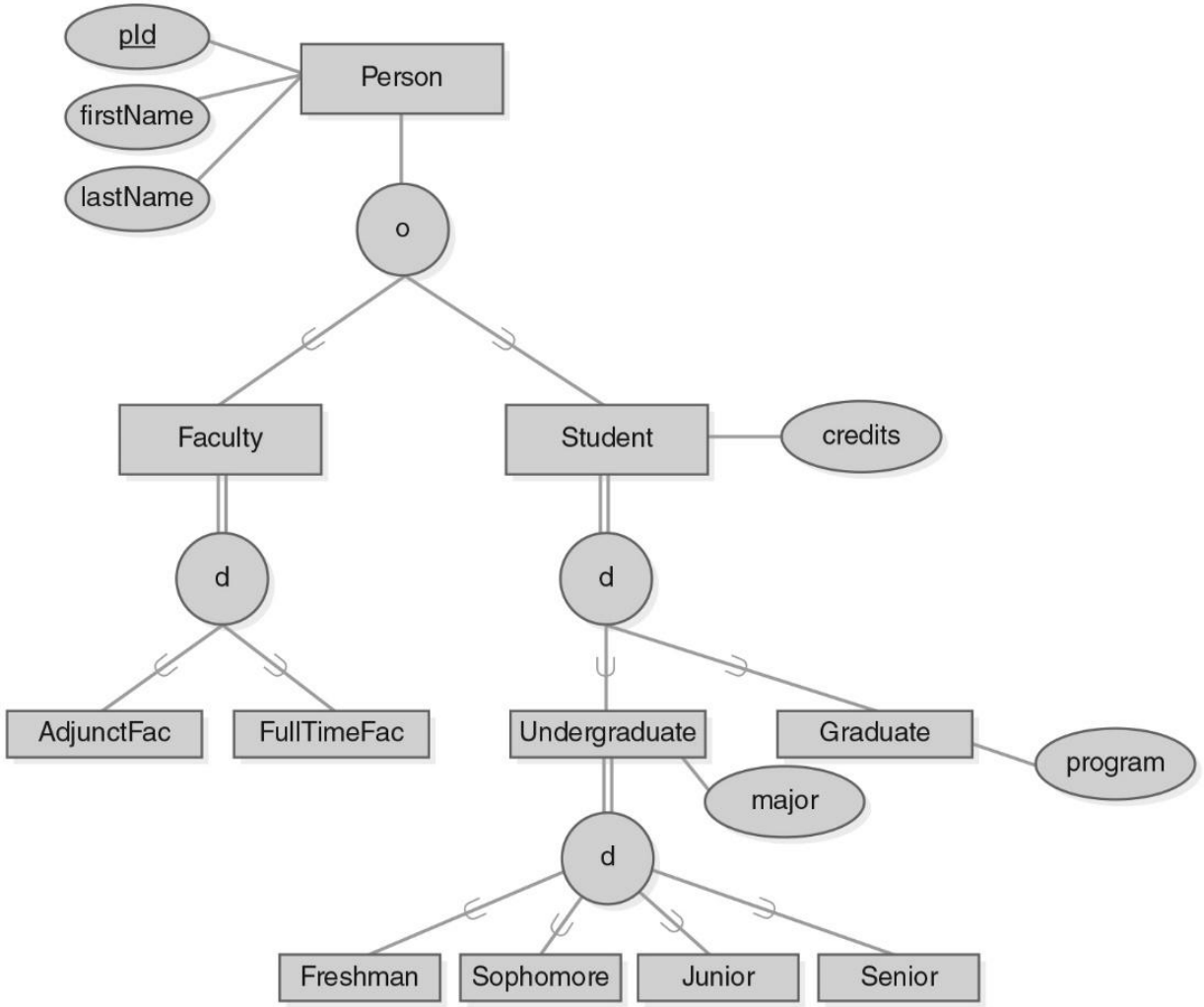

Figure 3.16 Disjointness and Participation Constraints

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- Specialization can be:
  - **Predicate-defined** – each subset has a defining predicate
  - **Attribute-defined** – the value of the **same attribute** is used in defining predicate for all subsets
  - **User-defined** – user responsible for identifying proper subset

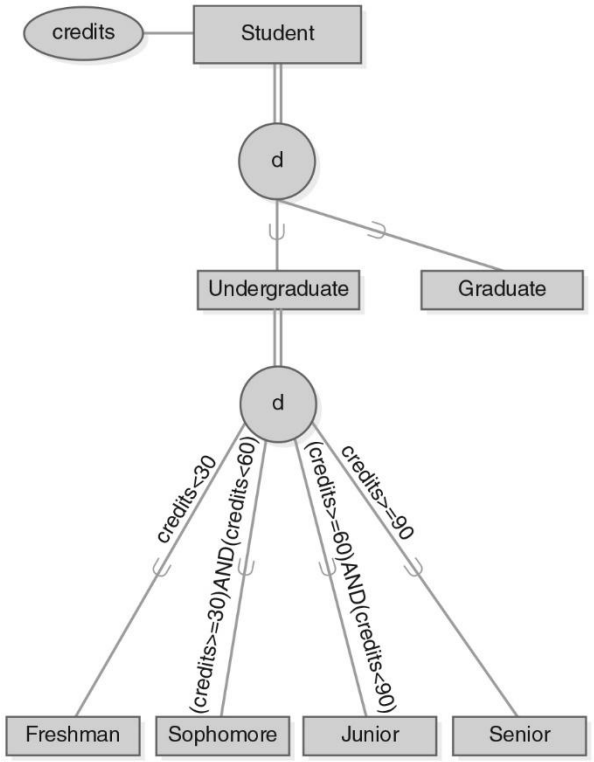

Figure 3.17A Predicate-defined Specialization

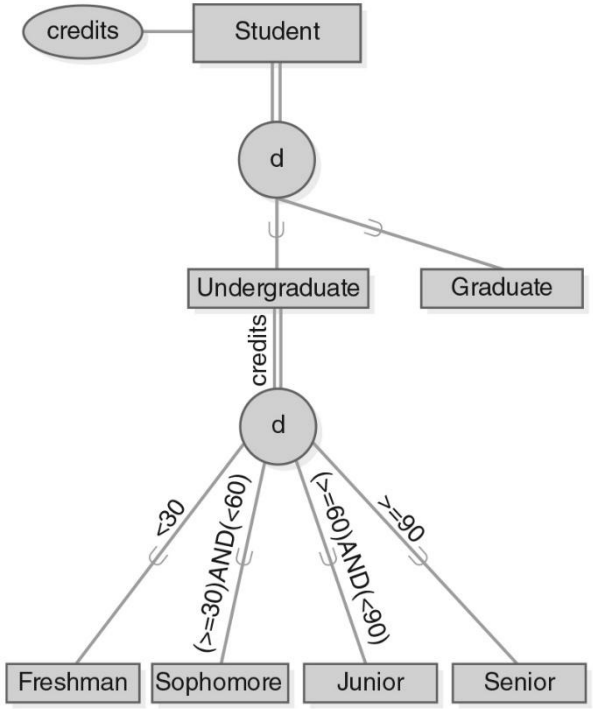

Figure 3.17B Attribute-defined Specialization

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Multiple Specializations & Multiple Inheritance

- Can have different specializations for the same set
- Figure 3.17(C) shows `Undergraduate` specialized by `year` and by `residence`. These are independent of each other

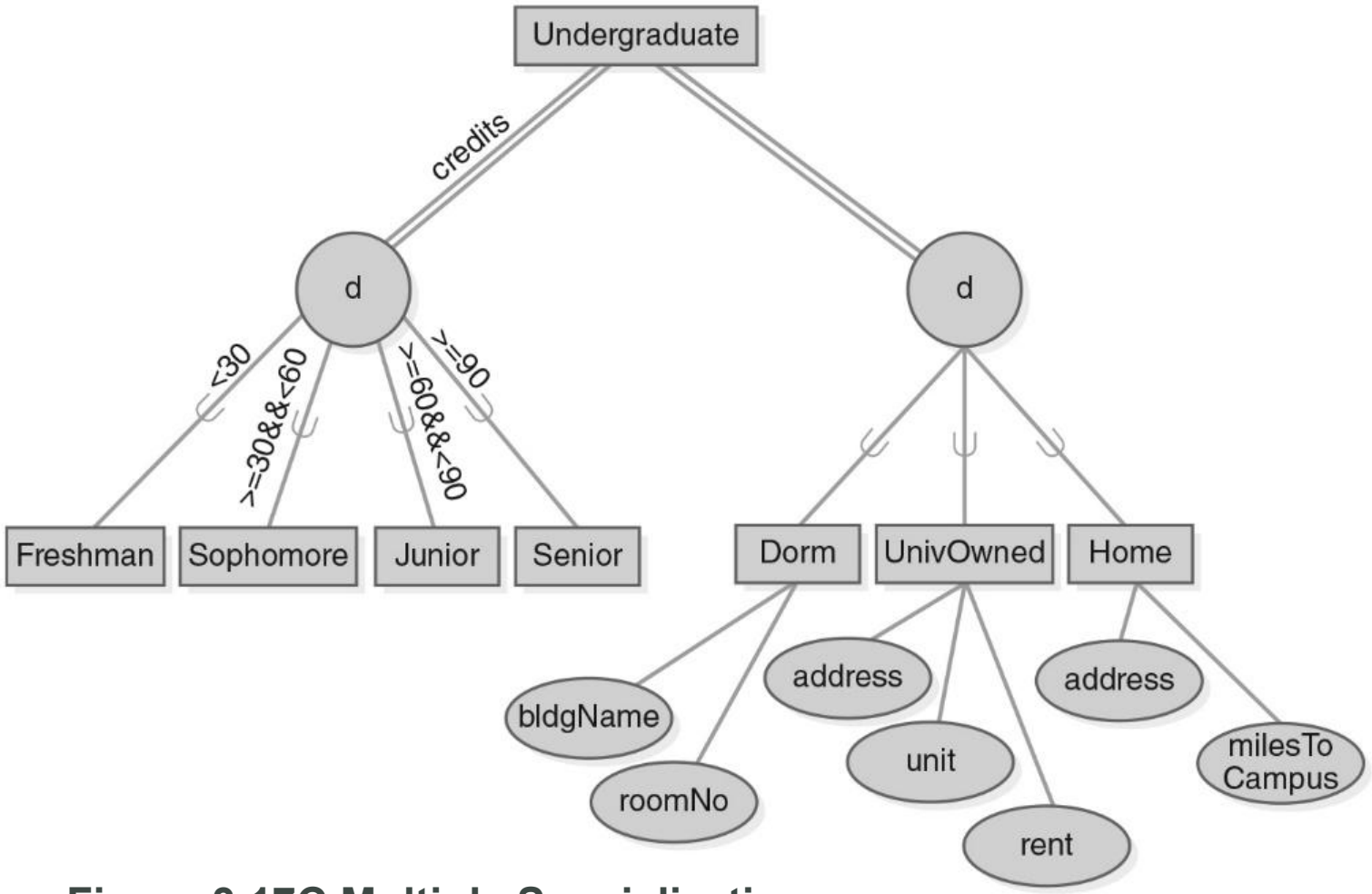

Figure 3.17C Multiple Specializations

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- Can have multiple inheritance – two or more supersets share same subset

  Ex. **Figure 3.18** shows `Teaching Assistant` as subset of both `Faculty` and `Student`

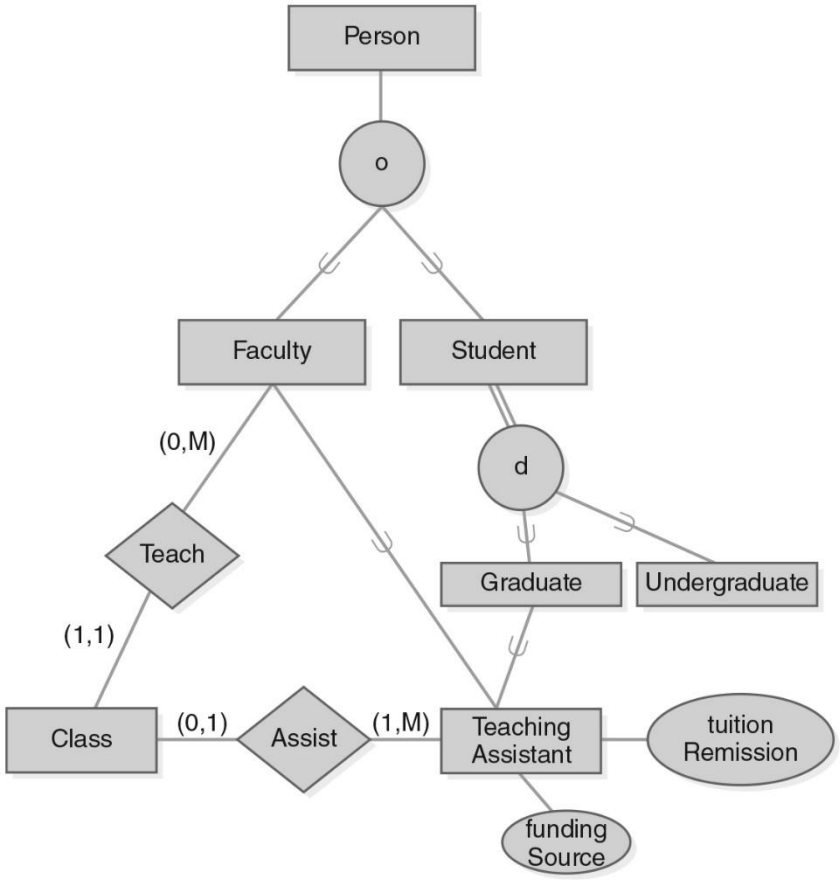

Figure 3.18 Multiple Inheritance

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Union or Category

- Subset related to a **collection** of supersets
- Each instance of subset belongs to one, not all, of the supersets
- Supersets form a union or category
- Ex. A `Sponsor` may be a `Team`, a `Department`, or a `Club`. See Figure 3.19A
- Each `Sponsor` entity instance is a member of one of these supersets, so `Sponsor` is a subset of the union of `Team`, `Department`, `Club`
- EER diagram – connect each superset to union circle, connect circle to subset, with subset symbol on line between circle and subset

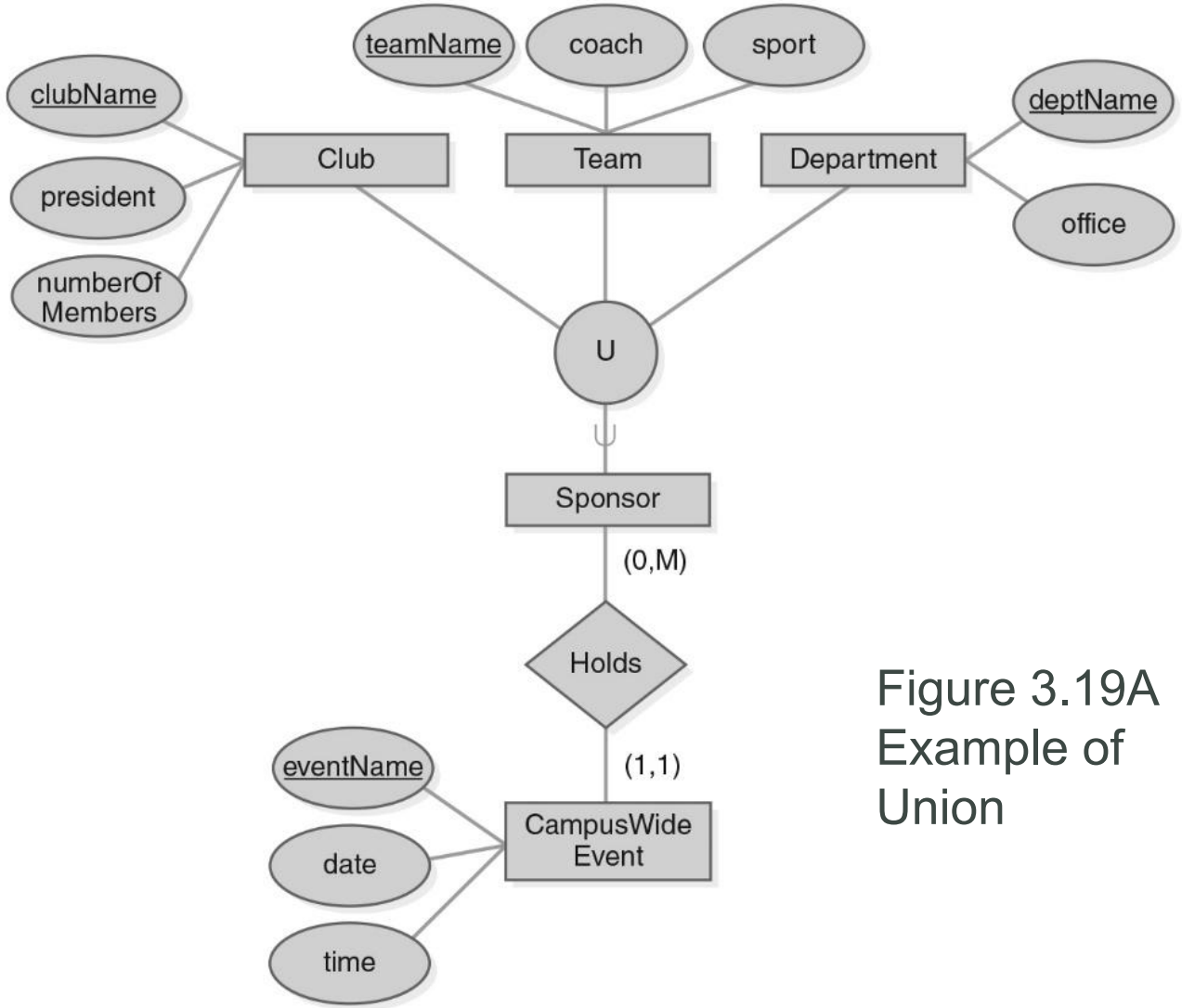

Figure 3.19A Example of Union

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Total and Partial Unions

- **Total category** – every member of the sets that make up the union must participate
  - Shown on EER by double line from union circle to subset
  - In Figure 3.19B every `Concert` or `Fair` must be a `Campus-Wide Event`
- **Partial category** – not every member of the sets must participate
  - Shown by single line
  - Not every `Club`, `Team`, or `Department` must be a `Sponsor`

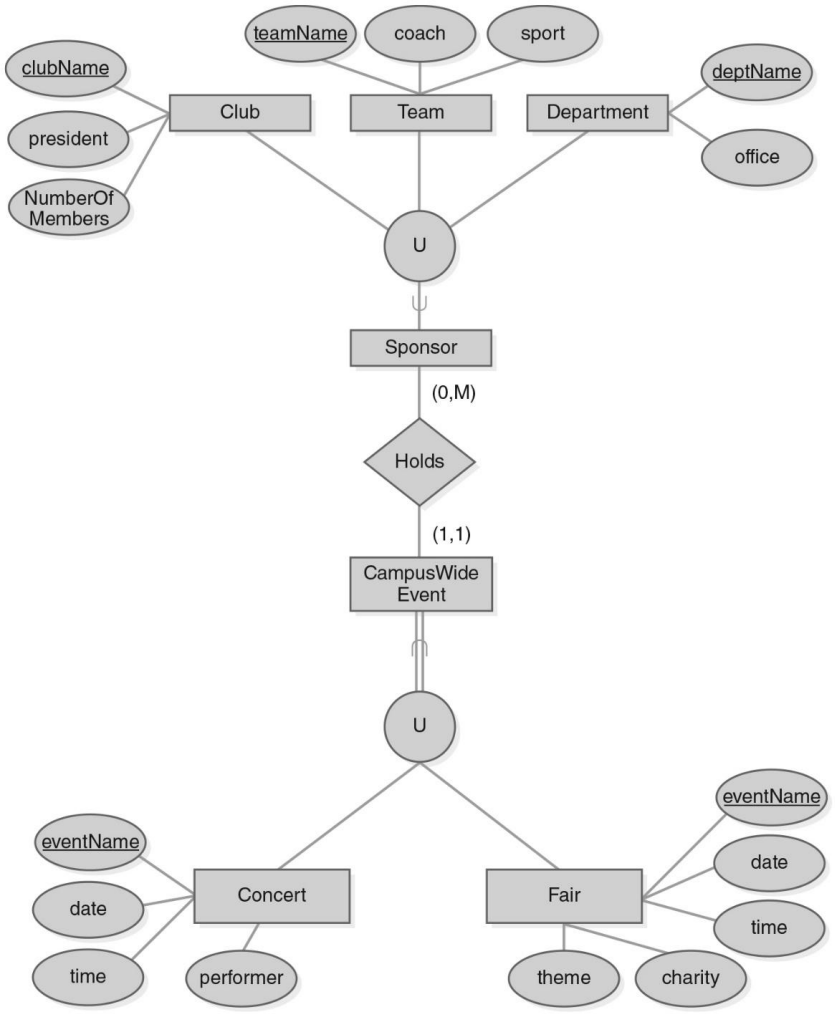

Figure 3.19B Partial vs. Total Union

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Total Union vs. Specialization

- Total union can often be replaced by hierarchy – See Figure 3.20
- A `Campus-Wide Event` must be either `Concert` or `Fair`
- Choose hierarchy representation if supersets have many common attributes
- Common attributes – `eventName`, `date`, `time`
- Add `type` to create attribute-type specialization

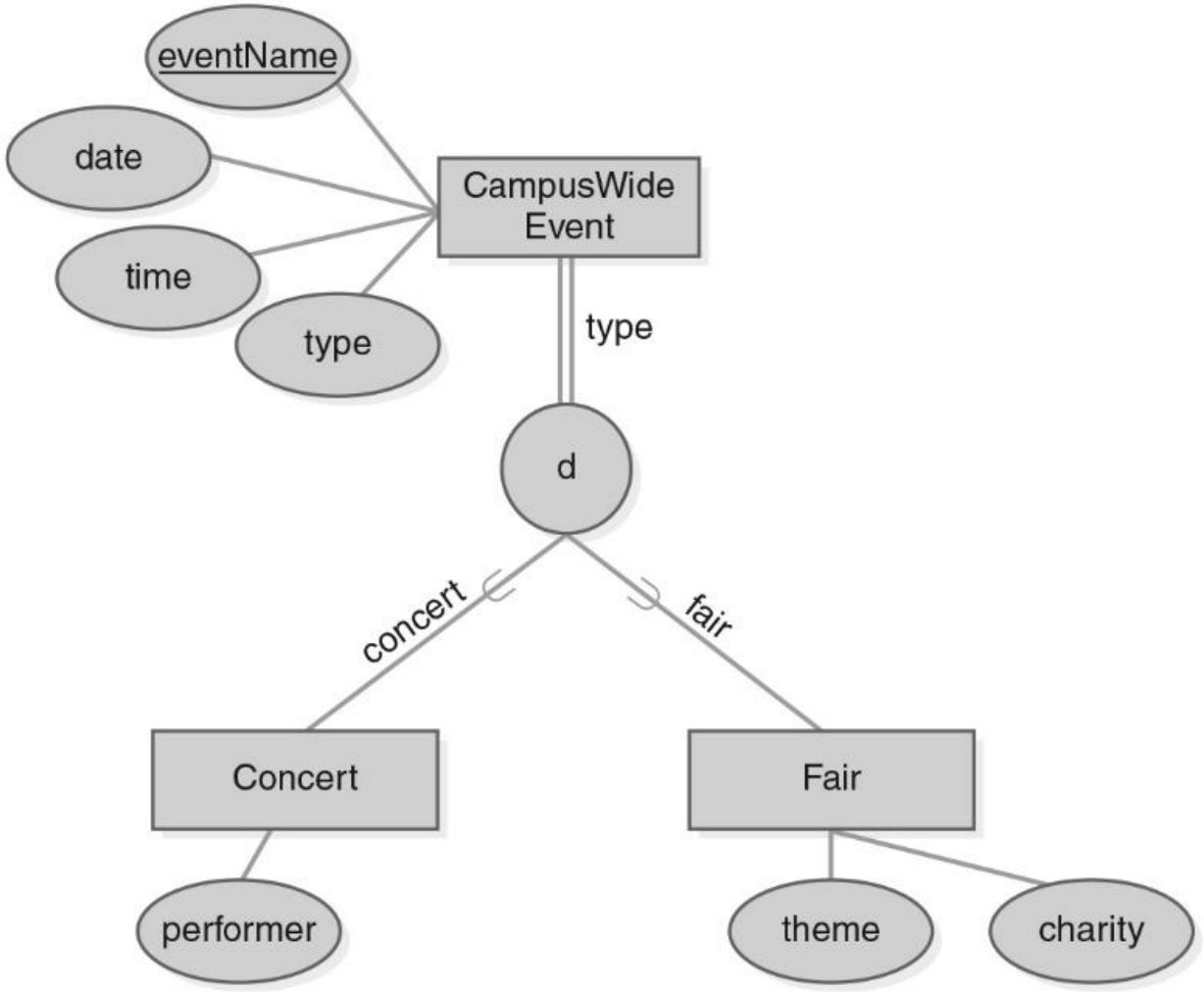

Figure 3.20 Total Union Replaced by Specialization/Generalization

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Sample EER Diagrams

- Extended `University` Example

see Figure 3.21

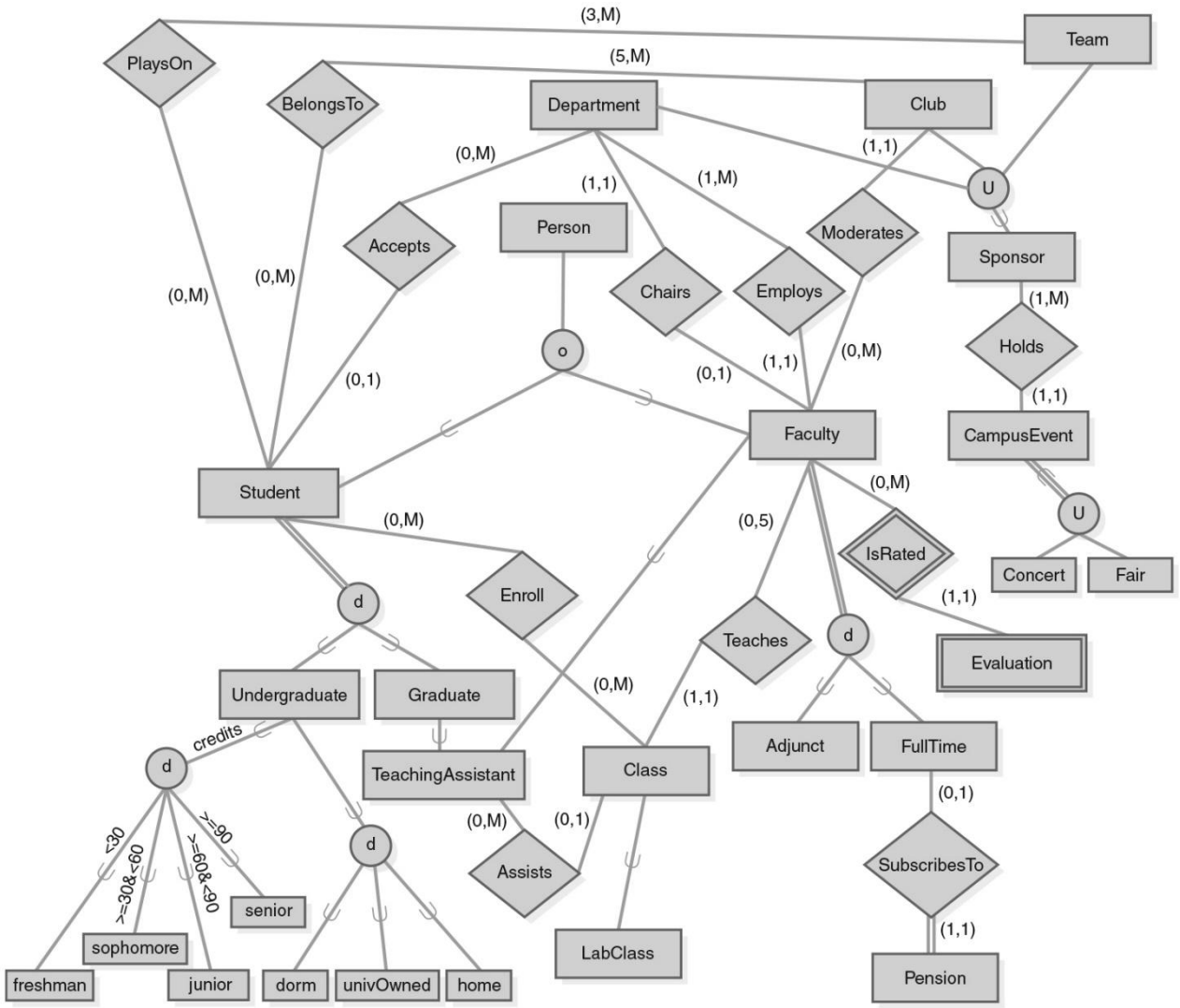

Figure 3.21 EER for `University Example`

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